Instead of only plotting the point,we can use annotate() to put a label and an arrow pointing to that point.

This can be done using annotate(). The syntax of annonate() is as bellow: ax.annotate(text, xy, xytext=None, xycoords='data', textcoords=None, arrowprops=None, **kwargs)

Parameter

text---> The text you want to display

xy----->Point being annoted-(x,y) coordinates

xytext----->position where the text should apper

xycoords----> Coordinate system used for xy; default is 'data'

textcoords-----> Coordinate System used for xytext

arrowprops------> Properties of the arrow connecting text to xy

**kwargs--------> other text properties such as fontsize, color, etc

```
import matplotlib.pyplot as plt

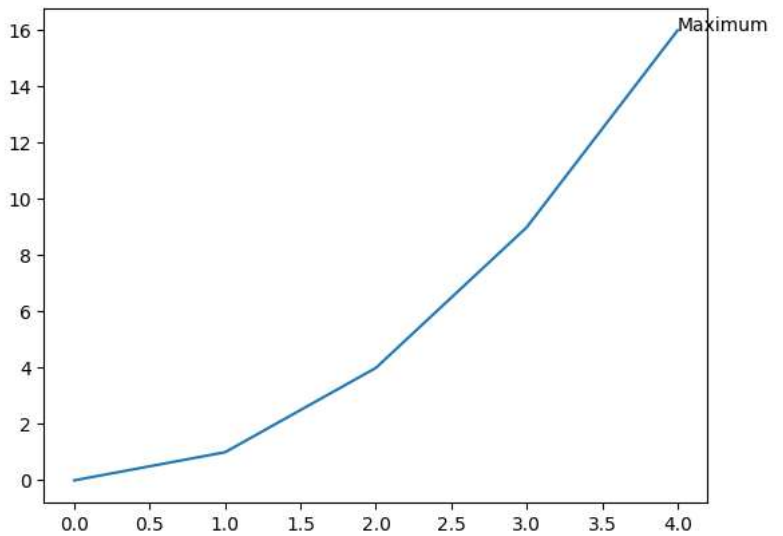
x = [0, 1, 2, 3, 4]
y = [0, 1, 4, 9, 16]

fig, ax = plt.subplots()

ax.plot(x, y)

ax.annotate('Maximum',
            xy=(4, 16))
            #xytext=(2, 12),
            #arrowprops=dict(arrowstyle='->'))

plt.show()
```




```
import matplotlib.pyplot as plt

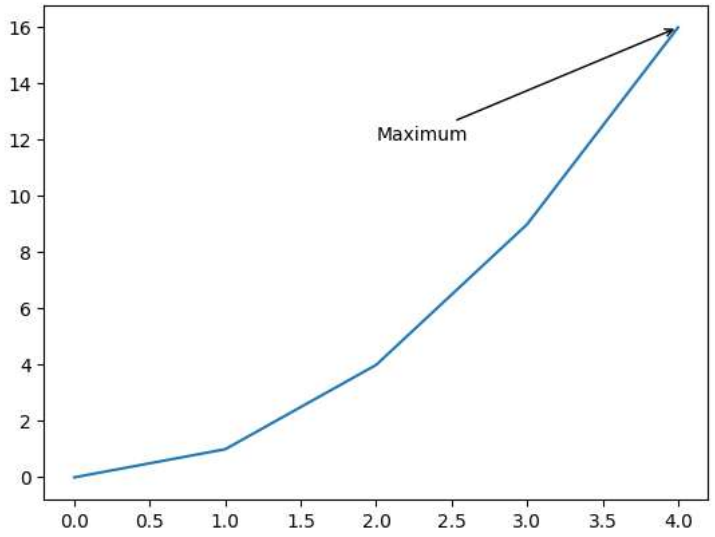
x = [0, 1, 2, 3, 4]
y = [0, 1, 4, 9, 16]

fig, ax = plt.subplots()

ax.plot(x, y)

ax.annotate('Maximum',
            xy=(4, 16),
            xytext=(2, 12),
            arrowprops=dict(arrowstyle='->'))

plt.show()
```



The annotate() statement tells Matplotlib: Put the word Maximum near (2,12) and draw an arrow toward (4,16). xy=(4,16) is the point that we want to identify. xy does not normally mean the location of the text.It means: "Point to which the annotation refers". xytext= (2,12) refers to the annotation text position. **What does arrowprops do?**

syntax:arrowprops=dict(arrowstyle='->')

tells Matplotlib to draw an arrow.

Another example marking the peak in an AC voltage waveform:

```
import numpy as np
import matplotlib.pyplot as plt

t = np.linspace(0, 20, 1000)
V = 325 * np.sin(2 * np.pi * t / 20)

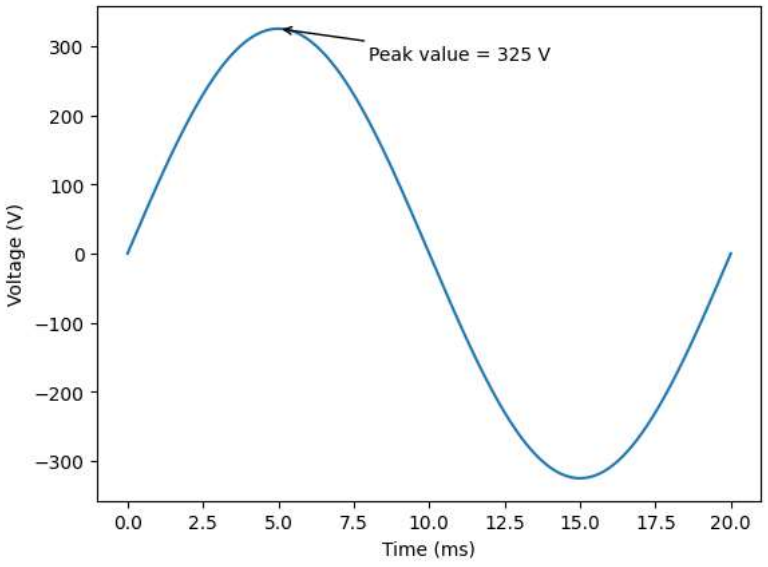
fig, ax = plt.subplots()

ax.plot(t, V)

ax.annotate('Peak value = 325 V',
            xy=(5, 325),
            xytext=(8, 280),
           arrowprops=dict(arrowstyle='->'))

ax.set_xlabel('Time (ms)')
ax.set_ylabel('Voltage (V)')

plt.show()
```



xycoords ⟶ How should Matplotlib understand the xy point?

textcoords ⟶ How should Matplotlib understand the xytext position?

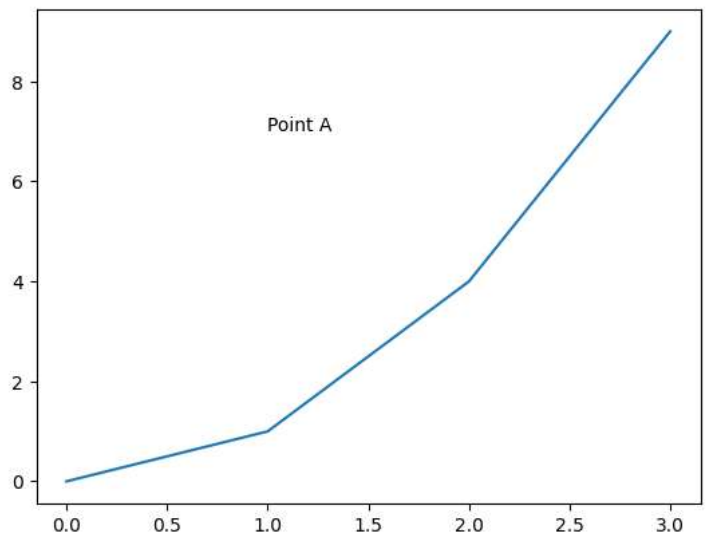
```
import matplotlib.pyplot as plt

fig, ax = plt.subplots()

ax.plot([0, 1, 2, 3], [0, 1, 4, 9])

ax.annotate(
    'Point A',
    xy=(2, 4),
    xycoords='data',
    xytext=(1, 7),
    textcoords='data'
)

plt.show()
```



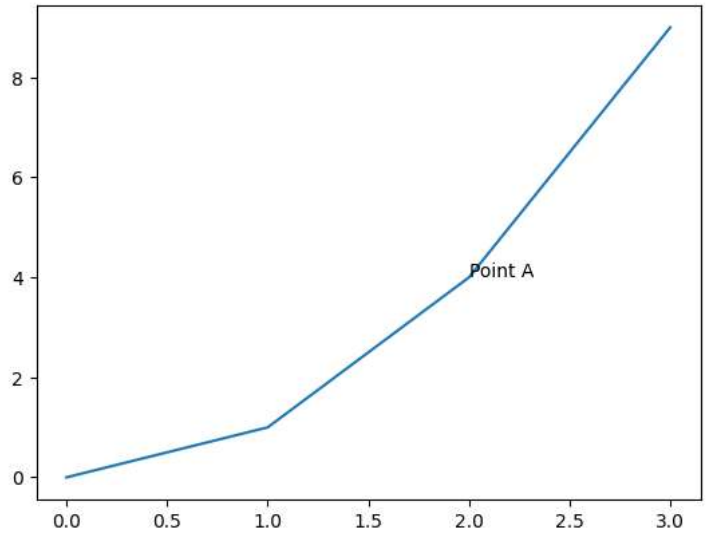

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots()

ax.plot([0, 1, 2, 3], [0, 1, 4, 9])

ax.annotate(
    'Point A',
    xy=(2, 4),
    xycoords='data',
  # xytext=(20,10),
   # textcoords='offset points'
)

plt.show()
```



xytext=(2,4) textcoords='data'

means: Put the text at x=2, y=4.

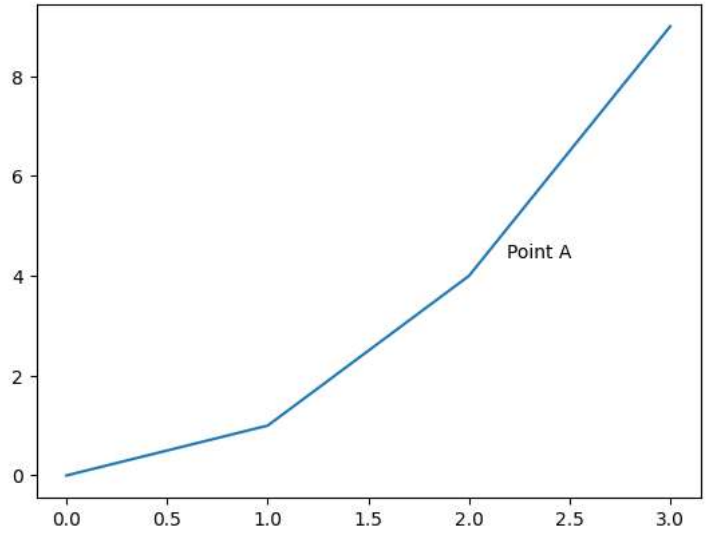
```
import matplotlib.pyplot as plt

fig, ax = plt.subplots()

ax.plot([0, 1, 2, 3], [0, 1, 4, 9])

ax.annotate(
    'Point A',
    xy=(2, 4),
    xycoords='data',
    xytext=(20,10),
    textcoords='offset points'
)

plt.show()
```



Close

```
xytext=(20,10)
textcoords='offset points'

means: Put the text 20 points right and 10 points up from xy.
```

xytext=(20,10) textcoords='offset points'

means: Put the text 20 points right and 10 points up from xy.